

Pacific Firestorm

Project overview, legal boundary, MVP, and repo structure

Purpose of This Kit

This kit is a complete Codex-agent handoff for building Pacific Firestorm: an original Godot 4 arcade aerial shooter inspired by the structure of classic vertical shooters, but with ultra-realistic top-down Pacific air-combat visuals.

Core rule: arcade gameplay, realistic visuals, original identity.

Game Goal

Build a vertical scrolling aerial shooter with simple, responsive arcade controls, enemy waves, power-ups, bosses, score chasing, and modern realistic presentation: aircraft, ocean, islands, smoke, fire, tracer rounds, explosions, weather, ships, and aircraft carriers.

Legal and creative boundary

- Do not use the 1942 name, Capcom branding, original sprites, sound effects, music, UI, enemy waves, or level layouts.
- Use the broad genre structure only: move, dodge, shoot, survive waves, collect power-ups, defeat bosses, chase score.
- Create original aircraft designs, original UI, original levels, original enemy wave timing, and original audio.

Recommended Technical Direction

Use a hybrid 3D/2D approach: Godot 4 with GDScript, a 3D world viewed by an orthographic Camera3D, and arcade gameplay locked to the X/Z plane.

Area	Recommendation
Engine	Godot 4.x
Language	GDScript first, C# only later if required
Camera	Camera3D, orthographic, top-down or slight tilt
Gameplay plane	X/Z plane with simplified hitboxes
Collision	Area3D and CollisionShape3D, fair arcade hitboxes
Data	JSON-driven enemies, waves, stages, weapons, power-ups

MVP Definition

The first real milestone is one complete playable stage, not a giant campaign. The MVP must prove that the game is fun before visual polish and extra content are expanded.

- Title screen and start game flow.
- Player aircraft with 8-direction arcade movement.
- Shooting, bullets, collisions, score, lives, HUD.

- Enemy fighters, enemy bullets, bomber, power-up, first boss.
- Scrolling ocean, simple clouds, basic explosions, game over, mission complete.

Build Strategy

Use Codex in phases. Give one phase prompt, run the Godot project, copy errors back to Codex, request minimum fixes, then continue. Avoid massive untested feature dumps.

1. Foundation
2. Main scene
3. Player movement
4. Shooting
5. Enemies
6. Collision/score
7. Lives/HUD
8. Wave spawner
9. Environment
10. Power-ups
11. Boss
12. Menus
13. Effects/audio
14. Stage 1
15. Visual realism
16. Performance/export

Repository Structure

```
PacificFirestorm/  
project.godot  
  
scenes/  
  main/Main.tscn  
  player/Player.tscn  
  enemies/EnemyFighter.tscn  
  enemies/EnemyBomber.tscn  
  enemies/BossBomber.tscn  
  bullets/PlayerBullet.tscn  
  bullets/EnemyBullet.tscn  
  bullets/Rocket.tscn  
  bullets/FlakBurst.tscn  
  powerups/Powerup.tscn  
  effects/Explosion.tscn  
  effects/SmokeTrail.tscn  
  effects/WaterSplash.tscn  
  effects/HitSpark.tscn  
  environment/Ocean.tscn  
  environment/IslandChunk.tscn  
  environment/CloudLayer.tscn  
  environment/Ship.tscn  
  environment/Carrier.tscn  
  ui/HUD.tscn  
  ui/TitleScreen.tscn  
  ui/PauseMenu.tscn  
  ui/GameOverScreen.tscn  
  ui/MissionCompleteScreen.tscn  
  
scripts/  
  main/main.gd  
  main/game_state.gd  
  main/stage_manager.gd  
  player/player.gd  
  player/player_weapon.gd  
  player/player_damage.gd  
  enemies/enemy_base.gd  
  enemies/enemy_spawner.gd  
  enemies/movement_patterns.gd  
  enemies/boss_controller.gd  
  bullets/bullet_base.gd  
  bullets/projectile_pool.gd  
  powerups/powerup.gd  
  powerups/powerup_spawner.gd  
  effects/explosion.gd  
  effects/effects_manager.gd  
  effects/screen_shake.gd
```

```
environment/ocean_scroller.gd
environment/cloud_scroller.gd
environment/environment_spawner.gd
ui/hud.gd
ui/title_screen.gd
ui/pause_menu.gd
ui/game_over_screen.gd
ui/mission_complete_screen.gd
systems/audio_manager.gd
systems/input_manager.gd
systems/save_manager.gd
systems/config_manager.gd

data/
  stages/stage_list.json
  stages/stage_01.json
  waves/stage_01_waves.json
  enemies/enemy_types.json
  weapons/weapon_types.json
  powerups/powerup_types.json

assets/
  models/aircraft/
  models/ships/
  models/islands/
  textures/ocean/
  textures/aircraft/
  textures/explosions/
  textures/clouds/
  textures/ui/
  materials/water/
  materials/aircraft/
  materials/smoke/
  materials/tracer/
  audio/music/
  audio/sfx/
  fonts/
  placeholders/

docs/
  game_design.md
  art_direction.md
  technical_plan.md
  controls.md
  codex_tasks.md
  known_issues.md
  legal_notes.md
  export_guide.md
  testing_checklist.md
  release_checklist.md

.github/pull_request_template.md
AGENTS.md
README.md
```